

Automated Compliance Auditing of Cloud Configurations

Bachelor's Thesis Proposal

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1 1 Introduction and Motivation

Cloud deployments change daily, and every change can silently violate regulatory requirements. Compliance audits in most organisations still happen quarterly and rely on manual spreadsheet reviews (Weber 2024). By the time an auditor discovers a misconfigured storage bucket, the violation may have persisted for months. Automated compliance auditing promises to close this gap by evaluating configuration snapshots against a machine-readable rule catalogue (Tanaka 2023). The proposed thesis develops and evaluates such an automated audit pipeline for infrastructure-as-code repositories.

2 2 Objectives and Work Packages

The overall objective is a working audit pipeline that flags non-compliant configurations before they reach production. The project is organised into the following work packages, which also serve as the guiding questions of the thesis:

1. How can the existing rule catalogue be translated into executable audit policies?
2. How can the audit engine be integrated into the continuous integration pipeline?
3. How can audit findings be exported into a report that compliance officers can act on?

3 3 Approach

The audit engine builds on an open-source policy language and evaluates configuration files at commit time (Osei 2025). A rule catalogue derived from a public cloud-security benchmark serves as ground truth. Detected violations are ranked by severity and written to a summary report. The approach is validated on a set of intentionally misconfigured example repositories.

4 4 Preliminary Chapter Structure

1. Introduction
2. Background and Related Work
3. Design of the Audit Pipeline
4. Implementation
5. Evaluation
6. Conclusion and Outlook

5 5 Timetable and Milestones

Month	Milestone
Month 1	Literature review complete
Month 2	Rule catalogue translated into policies
Month 3	Audit engine integrated into the pipeline
Month 4	Evaluation finished, thesis submitted

Osei, K. 2025. "Benchmarks for Automated Cloud Configuration Audits." *Journal of Example Systems Security*, ahead of print. <https://doi.org/10.xxxx/xxxx5>.

Tanaka, H. 2023. "Drift Detection in Infrastructure-as-Code Repositories." *Proceedings of the Example Conference on Cloud Computing*. <https://doi.org/10.xxxx/xxxx4>.

Weber, S. 2024. "Policy-as-Code for Continuous Compliance Auditing." *Journal of Example Cloud Engineering*, ahead of print. <https://doi.org/10.xxxx/xxxx3>.